Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Natural Language Processing**	**Course Code:**	**DS 5007**
Program:	**MS(Data Science)**	**Semester:**	**Fall 2021**
Duration:	**30 Minutes**	**Total Marks:**	**20**
Paper Date:	**24-Dec-21**	**Weight**	
Section:	**DS**	**Page(s):**	**3**
Exam:	**Quiz 4**		

Q1) Write steps involved in each of the following methods for relaiton extraction. [9 Marks]

a) Supervised Machine Learning

b) Bootstraping

c) Distant Supervision

Q2) Given the training data below:
Training Data:

- cat/NNS flying/VBG is/VBZ adventurous/JJ
- flying/JJ planes/NNS are/VBZ abundant/JJ
- I/PRP saw/VBZ Mary/NNP flying/VBG planes/NNS
- She/PRP planes/VBZ shelves/NNS

Calculate the probability of tag sequence ”JJ NNS VBZ JJ STOP” for the string ”flying planes are adventurous”. (Use bigram model with Laplace smooothing for transition probabilities). Show all calculations.

Solution

Number of uniques POS tags including STOP symbol = 7

number of unique words = |V| = 12

P(flying planes are adventurous, JJ NNS VBZ JJ STOP) =

P(flying|JJ)*P(planes|NNS)*P(are|VBZ)*P(adventurous|JJ) *

P(JJ|*) * P(NNS|JJ)*P(VBZ|NNS)*P(JJ|VBZ)*P(STOP|JJ)

= 1/3 * 2/4*1/4*1/3* (1+1)/(4+7) * (1+1/3+7) * (1+1)/(4+7)* (2+1)/(4+7)*(2+1)/(3+7)

= (1/3) * (2/4)*(1/4)*(1/3)* (2)/(11) * (2/10) * (2)/(11)* (3)/(11)*(3)/(10)

=7.5*10^{-6}